

RESEARCH ARTICLE

CANET: Quantized Neural Network Inference With 8-bit Carry-Aware Accumulator

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ABSTRACT Neural network quantization represents weights and activations with few bits, greatly reducing the overhead of multiplications. However, due to the recursive accumulation operations, high-precision accumulators are still required in multiply-accumulate (MAC) units to avoid overflow, incurring significant computational overhead. This constraint limits the efficient deployment of quantized NNs on resource-constrained platforms. To address this problem, we present a novel framework named CANET, which adapts the 8-bit quantized model to execute MAC operations with 8-bit accumulators. CANET not only employs 8-bit carry-aware accumulators to represent overflow data correctly, but also adaptively learns the optimal format per layer to minimize truncation errors. Meanwhile, a weight-oriented reordering method is developed to reduce the transfer length of the carry. CANET is evaluated on three networks in the ImageNet classification task, where comparable performance with state-of-the-art methods is realized. Finally, we implement the proposed architecture on a custom hardware platform, demonstrating a reduction of 40% in power and 49% in area compared with the MAC unit with 32-bit accumulators.

INDEX TERMS Convolutional neural network, quantization, efficient inference, low-precision accumulator.

I. INTRODUCTION

Neural network quantization facilitates the efficient deployment of deep neural networks (DNNs) on resource-constrained platforms, such as drones and Internet-of-Things (IoT) devices [1], [2]. During inference, MAC operations consisting of multiplications and accumulations dominate the arithmetic cost. Quantization successfully maps floating-point parameters to the low-precision fixed-point numbers, drastically reducing the cost of multiplications [3]. However, high-precision accumulators are necessary for deep feature computation, which requires the accumulation of thousands of products. The resulting computational overhead greatly limits the efficacy of quantization and prompts active exploration of inference based on low-precision accumulators [4].

Overflow is problematic when high-precision accumulators are replaced with low-precision counterparts. Because of overflow, out-of-range values are wrapped around to the

minimum representable value, resulting in severe data distortion [6]. Some impressive methods were proposed to reduce or eliminate the risk of overflow associated with low-precision accumulators [4], [6], [13], [29], [30]. Xie et al. [4] reduced the risk of overflow in low-precision accumulators by adaptively scaling weights and activations bit widths. Ni et al. [6] mitigated the impact of overflow on accuracy by cyclic activation layers. De Bruin et al. [13] employed a heuristic method to determine the appropriate parameter precision per layer for fixed-precision accumulators. However, frequent overflow imposes tight constraints on the parameter bit width, impairing the model accuracy in such methods.

This paper introduces a novel framework named CANET, which aims to employ 8-bit accumulators within the 8-bit quantization framework without compressing the bit width of weights and activations. CANET incorporates a carry-aware accumulator (C2A) with a fixed-point format to rectify wraparound data. The error introduced by C2A is quantized to truncation and swamping error. Additionally, an adaptive algorithm is designed for learning optimal fixed-point formats to minimize truncation error. Meanwhile,

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